Problem 1) $J(\alpha) = \int_{x_1}^{x_2} f(y(x), y'(x); x)\, dx$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial y'}\frac{\partial y'}{\partial \alpha} \right) dx \quad ; \quad \frac{\partial y}{\partial \alpha} = \eta(x) ; \frac{\partial y'}{\partial \alpha} = \frac{d\eta}{dx}$$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\eta(x) + \frac{\partial f}{\partial y'}\frac{d\eta(x)}{dx} \right) dx$$

Integration by parts gives us, $\int u\,dv = uv - \int v\,du$

$$\Rightarrow \int_{x_1}^{x_2} \frac{\partial f}{\partial y'}\frac{d\eta}{dx}\, dx = \underbrace{\frac{\partial f}{\partial y'}\eta(x)\Big|_{x_1}^{x_2}}_{=0 \text{ because } \eta(x_1) = \eta(x_2) = 0} - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\eta(x)\, dx$$

$$\Rightarrow \frac{\partial J}{\partial \alpha} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y}\eta(x) - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\eta(x) \right] dx$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \right] \eta(x)\, dx$$

$$\frac{\partial J}{\partial \alpha} = 0 \; ; \; \eta(x) \text{ is arbitrary} \Rightarrow \boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)} \quad \text{Euler's eqn.}$$

Problem 2) $L = T - U \; ; \; \frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = 0 \quad i = 1, 2, 3$

$\Rightarrow \frac{\partial (T-U)}{\partial x_i} - \frac{d}{dt}\frac{\partial (T-U)}{\partial \dot{x}_i} = 0$; Recall that for conservatie systems,

$U \equiv U(x_i) \; ; \; T \equiv T(\dot{x}_i) \Rightarrow \frac{\partial U}{\partial \dot{x}_i} = \frac{\partial T}{\partial x_i} = 0$

$$\left.\begin{aligned} &\Rightarrow -\frac{\partial U}{\partial x_i} = \frac{d}{dt}\frac{\partial T}{\partial \dot{x}_i} \\ &\text{For Conservative systems } F_i = -\frac{\partial U}{\partial x_i} \end{aligned}\right\} \Rightarrow \frac{d}{dt}\frac{\partial T}{\partial \dot{x}_i} = \frac{d}{dt}\frac{\partial}{\partial \dot{x}_i}\left(\sum_1^3 \tfrac{1}{2} m \dot{x}_j^2\right)$$

$$= \frac{d}{dt}(m\dot{x}_i) = \dot{P}_i$$

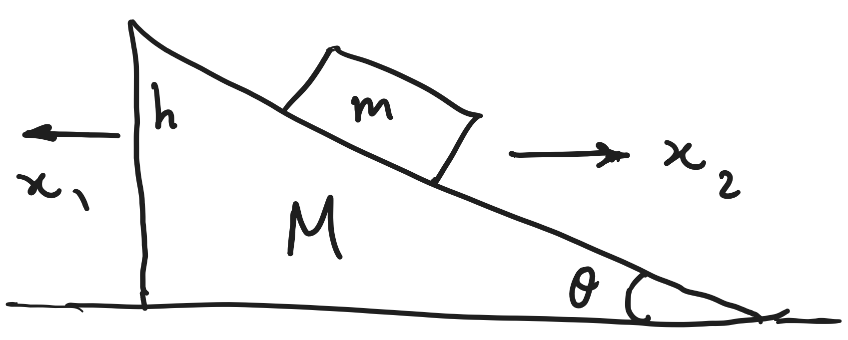
$$= F_i \Rightarrow \boxed{F_i = \dot{P}_i}$$

Problem 3) $L = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - mgz \; ; \; \frac{\partial L}{\partial x_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) = 0$

There is no x, y dependence; $\frac{\partial L}{\partial \dot{x}} = m\dot{x} \equiv$ const; $\frac{\partial L}{\partial \dot{y}} = m\dot{y} \equiv$ const.

Problem 4) x_1: The horizontal coordinate of M.

x_2: The horizontal coordinate of m.



$x_1 + x_2$: The relative horizontal distance between M, m.

$h = (x_1 + x_2)\tan\theta$ which is the height fallen by m.

$$\Rightarrow L = \frac{1}{2} M\dot{x}_1^2 + \frac{1}{2} m \left(\dot{x}_2^2 + \dot{h}^2\right) + mgh \quad \text{(The potential energy of } M \text{ is constant)}$$

The E-L eqns for x_1, x_2 yeild:
$$\begin{cases} M\ddot{x}_1 + m(\ddot{x}_1 + \ddot{x}_2)\tan^2\theta = mg\tan\theta \\ m\ddot{x}_2 + m(\ddot{x}_1 + \ddot{x}_2)\tan^2\theta = mg\tan\theta \end{cases}$$

$$\Rightarrow \boxed{\ddot{x}_1 = \frac{mg\sin\theta\cos\theta}{M + m\sin^2\theta}}$$

$M\ddot{x}_1 - m\ddot{x}_2 = 0$

$\Rightarrow \frac{d}{dt}\left(M\dot{x}_1 - m\dot{x}_2\right) = 0$

$\Rightarrow$ Linear momentum is constant.